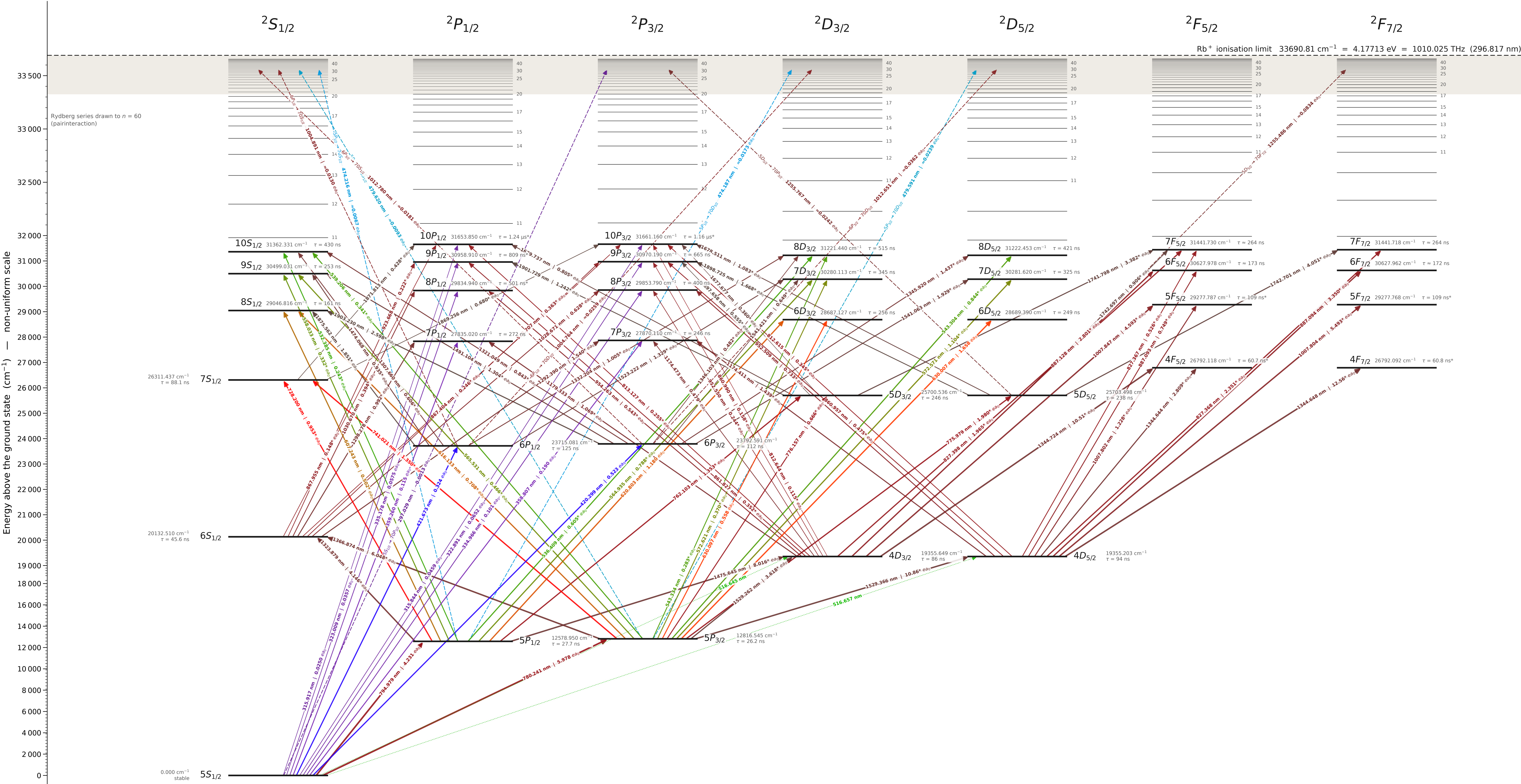
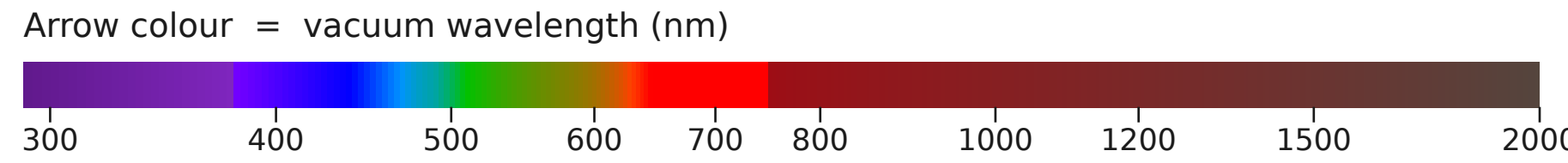


Rubidium-87

⁸⁷Rb (Rb I), nuclear spin $I = 3/2$. Energy levels and transitions below 2 μm : 36 levels, 94 lines.

Arrow label: vacuum wavelength λ | reduced dipole matrix element $\langle J||er||J' \rangle$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and lifetime (measured where a value exists, otherwise ARC, marked $\approx$).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\tilde{\nu}$ (cm^{-1})	$ \langle J er J' \rangle $ ($e a_0$)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Typical use
$5S_{1/2} \rightarrow 5P_{3/2}$	780.2412	780.0266	384.2305	12816.549	5.978	$3.811\text{e}+07$	6.065	100.0	D2: MOT, imaging, Rydberg lower leg
$5S_{1/2} \rightarrow 5P_{1/2}$	794.9788	794.7602	377.1075	12578.951	4.231	$3.610\text{e}+07$	5.746	100.0	D1: optical pumping, Λ / gray-molasses cooling
$5S_{1/2} \rightarrow 6P_{3/2}$	420.2989	420.1805	713.2839	23792.591	0.523	$1.866\text{e}+06$	1.421	21.1	blue leg of 420 + 1013 Rydberg excitation
$5S_{1/2} \rightarrow 6P_{1/2}$	421.6726	421.5539	710.9602	23715.081	0.324	$1.414\text{e}+06$	1.273	17.4	blue leg (421 + 1005)
$5P_{3/2} \rightarrow 5D_{5/2}$	775.9786	775.7651	386.3411	12886.953	1.980*	$2.833\text{e}+06$	0.667	66.8	780 + 776 ladder; 5S-5D two-photon at 778.1 nm
$5P_{1/2} \rightarrow 5D_{3/2}$	762.1030	761.8932	393.3752	13121.586	1.353*	$2.095\text{e}+06$	0.646	51.0	795 + 762 ladder
$5P_{3/2} \rightarrow 4D_{5/2}$	1529.3658	1528.9480	196.0240	6538.658	10.86*	$1.114\text{e}+07$	1.693	100.0	telecom C-band ladder
$5P_{3/2} \rightarrow 4D_{3/2}$	1529.2615	1528.8437	196.0374	6539.104	3.618*	$1.854\text{e}+06$	1.851	15.5	telecom C-band ladder
$5P_{1/2} \rightarrow 4D_{3/2}$	1475.6447	1475.2415	203.1603	6776.699	8.016*	$1.013\text{e}+07$	1.851	84.5	telecom S-band ladder
$5P_{3/2} \rightarrow 6S_{1/2}$	1366.8737	1366.5000	219.3271	7315.965	6.048*	$1.451\text{e}+07$	3.493	65.9	telecom E-band ladder
$5P_{1/2} \rightarrow 6S_{1/2}$	1323.8791	1323.5171	226.4500	7553.560	4.146*	$7.506\text{e}+06$	3.493	34.1	telecom O-band ladder
$5P_{3/2} \rightarrow 6D_{5/2}$	630.0068	629.8326	475.8559	15872.845	1.658	$3.712\text{e}+06$	0.639	77.9	red ladder
$5P_{3/2} \rightarrow 7S_{1/2}$	741.0211	740.8171	404.5667	13494.892	1.350*	$4.536\text{e}+06$	1.807	40.0	ladder to 7S
$6S_{1/2} \rightarrow 7P_{3/2}$	1292.3904	1292.0370	231.9674	7737.600	1.540*	$5.565\text{e}+05$	0.647	14.2	O-band, from 6S
$4D_{5/2} \rightarrow 4F_{7/2}$	1344.6483	1344.2807	222.9523	7436.889	12.56*	$1.643\text{e}+07$	2.618	98.2	strongest line of the diagram
$6P_{3/2} \rightarrow 7D_{5/2}$	1541.0626	1540.6416	194.5362	6489.029	1.929*	$3.433\text{e}+05$	0.490	11.5	telecom C-band, from 6P
$6P_{3/2} \rightarrow 8S_{1/2}$	1903.2303	1902.7107	157.5177	5254.225	2.598*	$9.918\text{e}+05$	0.989	16.0	longest 6P line below 2 μm

λ and ν from NIST level energies (centre of gravity, no hyperfine structure); λ air uses standard air (Giddor).
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$. Branch: share of the upper level's decays that takes this line.

Rydberg excitation wavelengths (vacuum, nm)

Lower $\rightarrow$ series	$n = 30$	$n = 40$	$n = 50$	$n = 70$	$n = 100$	$n \rightarrow \infty$	$d_{\nu 0}$ pairint.	$d_{\nu 0}$ ARC
$5S_{1/2} \rightarrow nP_{3/2}$	298.113	297.510	297.247	297.029	296.918	296.817	0.0013	0.0023
$5P_{1/2} \rightarrow nS_{1/2}$	477.099	475.483	474.788	474.216	473.927	473.667	0.0067	0.0067
$5P_{1/2} \rightarrow nD_{3/2}$	476.682	475.318	474.707	474.187	473.917	473.667	0.0173	0.0166
$5P_{3/2} \rightarrow nS_{1/2}$	482.570	480.916	480.205	479.620	479.324	479.059	0.0093	0.0095
$5P_{3/2} \rightarrow nD_{3/2}$	482.143	480.747	480.122	479.591	479.315	479.059	0.0239	0.0230
$4D_{5/2} \rightarrow nP_{3/2}$	704.765	701.404	699.946	698.736	698.121	697.564	0.0100	0.0092
$4D_{5/2} \rightarrow nF_{7/2}$	703.548	700.913	699.701	698.649	698.092	697.564	0.0559	0.0582
$6S_{1/2} \rightarrow nP_{3/2}$	745.611	741.850	740.219	738.866	738.178	737.556	0.0055	0.0061
$6P_{1/2} \rightarrow nS_{1/2}$	1017.930	1010.598	1007.464	1004.891	1003.596	1002.433	0.0130	0.0130
$6P_{1/2} \rightarrow nD_{3/2}$	1016.033	1009.855	1007.100	1004.764	1003.554	1002.433	0.0258	0.0253
$6P_{3/2} \rightarrow nS_{1/2}$	1026.025	1018.576	1015.393	1012.780	1011.464	1010.283	0.0181	0.0181
$6P_{3/2} \rightarrow nD_{5/2}$	1024.097	1017.821	1015.023	1012.651	1011.421	1010.283	0.0361	0.0354
$5D_{3/2} \rightarrow nP_{3/2}$	1275.375	1264.411	1259.681	1255.767	1253.782	1251.986	0.0242	0.0236
$5D_{5/2} \rightarrow nF_{7/2}$	1271.394	1262.817	1258.887	1255.486	1253.687	1251.986	0.0834	0.0846
$7S_{1/2} \rightarrow nP_{3/2}$	1382.573	1369.698	1364.148	1359.560	1357.233	1355.129	0.0114	0.0118
$4F_{7/2} \rightarrow nD_{5/2}$	1478.152	1465.114	1459.322	1454.424	1451.889	1449.545	0.0186	0.0186
$7P_{3/2} \rightarrow nS_{1/2}$	1764.032	1742.128	1732.838	1725.240	1721.425	1718.006	0.0291	0.0291
$7P_{3/2} \rightarrow nD_{5/2}$	1758.339	1739.920	1731.758	1724.865	1721.300	1718.006	0.0491	0.0485

Rydberg term energies from pairinteraction, lower level from NIST. $d_{\nu 0} = |\langle J||er||J' \rangle|$ to $n = 70$ in $e a_0$ (calculated).
 $S \rightarrow nP$ values come from a near-cancelling integral: treat them as an order of magnitude.

⁸⁷Rb hyperfine structure ($I = 3/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$5S_{1/2}$	3417.341305	-	F=1: -4271.68 F=2: +2563.01
$5P_{1/2}$	406.5(3)	-	F=1: -508.10 F=2: +304.86
$5P_{3/2}$	84.720(3)	12.4970(20)	F=0: -302.08 F=1: -229.86 F=2: -72.91 F=3: +193.74
$4D_{5/2}$	-16.786(10)	3.82(16)	F=1: +90.80 F=2: +54.17 F=3: +2.10 F=4: -61.99
$4D_{3/2}$	24.75(12)	2.19(11)	F=0: -90.08 F=1: -67.52 F=2: -20.20 F=3: +56.23
$6S_{1/2}$	807.35(4)	-	F=1: -1009.19 F=2: +605.51
$6P_{1/2}$	132.569(9)	-	F=1: -165.71 F=2: +99.43
$6P_{3/2}$	27.706(11)	3.97(3)	F=0: -98.93 F=1: -75.20 F=2: -23.76 F=3: +63.33
$5D_{3/2}$	14.4303(5)	0.9320(17)	F=0: -52.95 F=1: -39.45 F=2: -11.52 F=3: +32.70
$5D_{5/2}$	-7.4605(3)	1.2713(20)	F=1: +40.06 F=2: +24.12 F=3: +1.17 F=4: -27.66
$7S_{1/2}$	319.7500(20)	-	F=1: -399.69 F=2: +239.81
$7P_{1/2}$	59.92(9)	-	F=1: -74.90 F=2: +44.94
$7P_{3/2}$	12.570(10)	1.762(16)	F=0: -44.94 F=1: -34.13 F=2: -10.75 F=3: +28.72
$6D_{3/2}$	7.84(5)	0.53(6)	F=0: -28.74 F=1: -21.43 F=2: -6.28 F=3: +17.77
$6D_{5/2}$	-3.61(6)	-0.20(20)	F=1: +18.81 F=2: +11.75 F=3: +1.01 F=4: -13.59
$7D_{3/2}$	4.53(3)	0.26(4)	F=0: -16.66 F=1: -12.39 F=2: -3.59 F=3: +10.26
$7D_{5/2}$	-1.98(28)	-	F=1: +10.39 F=2: +6.43 F=3: +0.49 F=4: -7.42

A, B: magnetic-dipole and electric-quadrupole constants (measured; see data file for sources).